ACADEMIC CITY UNIVERSITY

FACULTY OF COMPUTATIONAL SCIENCES AND INFORMATICS

END OF SECOND SEMESTER EXAMINATION – 2026
CS4241-Introduction to Artificial Intelligence
QUESTION PAPER

Time: 10 Days **Maximum Marks: 60** **Credits Hours : 3**

Examination Date : 15th April 2026 Lecturer: Godwin N. Danso Student's Roll No: ___

Instructions :

This is a project-based examination. You are required to answer all questions. Your solutions should incorporate the methodologies, technologies, and criteria specified. Your project will demonstrate your ability to apply Artificial Intelligence concepts , theories and tools.

Submit your solution to github and deploy the solution to the cloud. The repository name should be ai_***index number*** . where ***index number*** is your student index number. Also include your name and index number in the readme file and every file.

You are to submit unique work. Any duplicated work , coping of someone's work will result in low or no marks

You are to share both your GitHub (repository) ,the deployed solutions(URL) links and any other documentations with the email: godwin.danso@acity.edu.gh.

The email title or subject should be:CS4241-Introduction to Artificial Intelligence-2026:[insert your index and name]

Add or invite godwin.danso@acity.edu.gh or GodwinDansoAcity as a GitHub collaborator.
Failure to do so will result in getting nothing for your exams.

Design and implement a RAG application that allows users to chat with your app

- You must include:
  - A **video walkthrough (not more than 2 minutes)** explaining your design decisions and the solutions
  - **Manual experiment logs** (not AI-generated summaries)

Important Constraint:
You are **NOT allowed** to use end-to-end frameworks such as **LangChain, LlamaIndex, or pre-built RAG pipelines**.
You must implement **core RAG components manually** (retrieval, chunking, embedding logic, prompt construction).

MAIN QUESTION: DESIGN A RAG System chatbot (60 MARKS)

You are tasked to build a RAG **AI chat assistant** for **Academic City**.

- **Use the following dataset for the project:**
  - https://github.com/GodwinDansoAcity/acitydataset/blob/main/Ghana_Election_Result.csv
  - https://mofep.gov.gh/sites/default/files/budget-statements/2025-Budget-Statement-and-Economic-Policy_v4.pdf

PART A: DATA ENGINEERING & PREPARATION (4 MARKS)

Requirements:

1. Perform:
   - Data cleaning
   - Chunking strategy design (**justify chunk size and overlap**)
2. Implement **your chunking strategies** .

Deliverables:

- Code implementation
- Justification of design decisions
- Comparative analysis of chunking impact on retrieval quality

PART B: CUSTOM RETRIEVAL SYSTEM (6 MARKS)

Requirements:

1. Implement your own:
   - Embedding pipeline (e.g., sentence transformers or OpenAI embeddings)
   - Vector storage (FAISS / Chroma / custom structure)
2. Implement:
   - Top-k retrieval
   - Similarity scoring
3. Extend retrieval with ONE of the following:
   - Query expansion
   - Re-ranking

- Hybrid search (keyword + vector)

Critical Task:

- Show **failure cases** where retrieval returns irrelevant results
- Propose and implement a fix

PART C: PROMPT ENGINEERING & GENERATION (4 MARKS)
Requirements:

1. Design a **prompt template** that:
   - Injects retrieved context
   - Handles hallucination control
2. Implement:
   - Context window management (truncate, rank, or filter chunks)
3. Conduct experiments:
   - Same query with different prompts
   - Analyze output differences

Deliverables:

- Prompt design iterations
- Evidence of improvement

PART D: FULL RAG PIPELINE IMPLEMENTATION (10 MARKS)
Requirements:
Build a complete pipeline:
User Query → Retrieval → Context Selection → Prompt → LLM → Response
Additional Constraints:

- Implement **logging at each stage**
- Display:
  - Retrieved documents
  - Similarity scores
  - Final prompt sent to LLM

PART E: CRITICAL EVALUATION & ADVERSARIAL TESTING (6 MARKS)
Requirements:

1. Design **2 adversarial queries**, such as:
   - Ambiguous queries

- Misleading or incomplete queries

2. Evaluate:
   - Accuracy
   - Hallucination rate
   - Response consistency
3. Compare:
   - Your RAG system vs **pure LLM (no retrieval)**

You must provide **evidence-based comparison**, not opinion.

PART F: ARCHITECTURE & SYSTEM DESIGN (8 MARKS)

Requirements:

- Draw and explain your architecture:
  - Data flow
  - Components interaction
- Justify:
  - Why your design is suitable for the chosen domain

PART G: INNOVATION COMPONENT (6 MARKS)

Introduce **ONE novel feature**, such as:

- Feedback loop for improving retrieval
- Memory-based RAG
- Multi-step reasoning (chain-of-thought orchestration)
- Domain-specific scoring function

FINAL DELIVERABLES (16 marks)

i. Application (4 Marks)

- The GitHub solution
- Simple UI (Streamlit / Next.js / Flask)
- Features:
  - Query input
  - Display retrieved chunks

- Show final response

ii. A **video walkthrough (not more than 2 minutes)** explaining your design decisions and the solutions[4 marks]

iii. **Manual experiment logs** (not AI-generated summaries) [4 marks]

iv. Documentation(detailed not simple AI-generated summaries)[4 marks]

v. The architecture (marks, see part F)

THE END